

Handling Date Values For Database Operations

05 September 2026 12:04

There are 2 date class available:

`java.util.Date` → represents both date and time

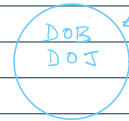
`java.sql.Date` → class ment for database operations

class represents only Date values
but not time.

`java.sql.Time`

`java.sql.Timestamp`

emp-id ← int
ename ← String
esalary ← double
eaddress ← String



String DOB = "28-05-1980";
↑
not recommended.

Handling Date values for Database operations

Sometimes as the part of programming requirement, we have to insert and retrieve Date like DOB, DOJ, DOP... with respect to database. It's not recommended to maintain date values in the form of String because comparison will become difficult.

In Java we have two date classes:

1. `java.util.Date`
2. `java.sql.Date`

`java.sql.Date` is specially designed class for handling Date values wrt database. Other than database operations; if we want to represent Date in our java program then we should go for `java.util.Date`.

`Java.util.Date` can represent both date and time whereas `java.sql.Date` represents only Date but not time.

```
package MyJDBCOne;
import java.sql.Date;
public class TestDateAndTime {
    public static void main(String[] args) {
        java.util.Date ddate = new java.util.Date();
        long l = ddate.getTime();
        Date udate = new Date(l);
        System.out.println("java.util.Date = " + ddate);
        System.out.println("java.sql.Date = " + udate);
    }
}
```

Output:

```
java.util.Date = Sat Sep 05 12:30:25 IST 2026
java.sql.Date = 2026-09-05
```

Note: In SQL package, Time class is available to represent time values and Timestamp classes available to represent both date and time.

Inserting date values into Database:

Oracle: dd-mmm-yy

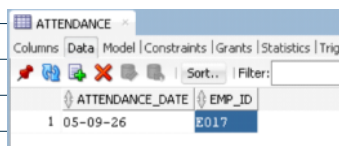
MySQL: yyyy-mm-dd

Statement: Java programmer is responsible to convert date values to respected DB date types.

```
PreparedStatement pst = ...
Java.util.Date attendanceDt = new java.util.Date();
long t = attendanceDt.getTime();
Java.sql.Date sqlDt = new java.sql.Date(t);
pst.setDate(1, sqlDt);
```

```
package MyJDBCOne;
import java.sql.Connection;
import java.sql.Date;
import java.sql.PreparedStatement;
import java.sql.SQLException;
import java.util.Scanner;
public class DateInsertDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Connection con = ConnectionObject.getConnection();
        String sql = "INSERT INTO ATTENDANCE VALUES(?,?)";
        try {
            PreparedStatement pst = con.prepareStatement(sql);
            System.out.println("Enter employee id: ");
            String empid = sc.next();
            long l = new java.util.Date().getTime();
            Date sdate = new Date(l);
            pst.setDate(1, sdate);
            pst.setString(2, empid);
            pst.executeUpdate();
            System.out.println("Record inserted");
        } catch (SQLException e) {
            e.printStackTrace();
        }
        sc.close();
    }
}
```

Output:



ATTENDANCE_DATE	EMP_ID
05-09-26	E017

